
Ayuda Home Health

Proposal

to be presented on 12/10/2025

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1. INTRODUCTION

Ayuda Home Care Agency currently uses paper recording and spreadsheets to manage patient intake, caregiver onboarding, CNA certification tracking, timesheet processing, and care plan documentation. This manual system is inefficient and causes delayed approvals, inaccurate records, increased administrative workload, and hinders audit preparation.

This project is designed to implement a centralized digital management system that consolidates major workflows. The system will support patient and caregiver registration, digital care plan creation, automated certification and compliance tracking, electronic document storage, while ensuring secure role-based accesses and complete audit logging. The proposed solution is expected to deliver substantial benefits to Ayuda Home Care. Benefits include improved accuracy and accessibility of records, reduced administrative burden through digitization, and automation. Integrated regulatory compliance, faster approval, and reporting processes. By modernizing its information infrastructure, the agency will strengthen its capacity to provide efficient and compliant home care services.

1.1 PROBLEM STATEMENT

- Ayuda Home Care Agency currently relies on a paper and spreadsheet driven system to manage its operational processes. Patient intake, caregiver onboarding, timesheets, and care plan documentation all suffer from inconsistent record keeping and delayed approvals. Difficulty monitoring certification deadlines and frequent data entry errors combined with the absence of a centralized information system also complicates communication with doctors and Medicaid reviewers making it difficult to retrieve documents and prepare materials during audits
- Administrative staff face multiple inefficiencies due to manual data entry and physical file management. Nurses must create and update care plans by hand limiting visibility to relevant stakeholders and causing difficulty in maintaining accurate version control. Care givers may have confusion regarding certification deadlines and delayed notification. Patients and families are unable to track their intake status, leading to uncertainty and repetitive inquiries processed by administrative staff. Slow or delayed information queries increase the risk of noncompliance with auditors and government agencies and lead to operational disruptions.
- The collective impact of these problems hinders organizational efficiency and introduces risks with regulatory bodies. Slow service delivery restricts the agency's ability to scale operations. Staff lack a centralized method to track insurance status or required

documentation. The distribution of documents under Ayuda's current system complicates insurance audit preparation and limits informational accessibility at an organizational level.

- Ayuda requires a centralized HIPAA compliant digital management system capable of consolidating patient and caregiver records, improving organizational level information access. This system will automate certification and approval workflows, digitize fax and timesheet processing, and generate audit ready reports. Introducing this system will reduce administrative strain and improve data accuracy while supporting long term operational stability.

1.2 BACKGROUND

Ayuda Home Care Agency provides home services to senior citizens across the Denver Metro Area through Medicare and Medicaid programs. The agency supports two types of caregivers. Independent caregivers and familial caregivers, each with distinct roles and certification requirements and service types. Familial caregivers who typically stay with the patient, receive compensation for providing care, and may pursue a CNA certification with agency support. They function under a four-month grace period during which they may deliver services and receive payment while completing the required certification. The agency's services include homemaking assistance, health maintenance assistance, personal care, and medically centered home health services provided by a CNA certified caregiver.

To manage these operations administrations are responsible for patient registration, caregiver onboarding, verification of credentials, tracking of familial caregivers CNA certification progress, logging caregiver hours, preparing payroll reports and documenting any medical changes affecting patient care plans. A certified nurse crafts individualized patient care plans, conducts necessary home visits and records observations and care plan updates. All of these activities require accurate record keeping on time updates and reliable coordination with administrative staff, caregivers, nurses, and external medical partners.

1.3 NEEDS STATEMENT

Ayuda Home Care Agency's current paper and spreadsheet-driven information management system is no longer adequate to support the operational, regulatory, and documentation requirements of a modern home care provider. The agency must maintain detailed records of elderly patients, caregivers, CNA certifications status, nurse visits, time reporting, and Medicaid approval workflows. Under the existing system, these records are stored in filing cabinets,

handwritten forms, and isolated spreadsheets. This leads to frequent delays, misplaced documents, errors, and inconsistent access to critical information. Nurses must create and update care plans by hand, and caregivers submit paper timesheets requiring manual transcription. Administrators must retrieve documents across multiple folders when responding to Medicaid or Medicare audits. This undermines the agency's efficiency.

The need for this project is further supported by demographic and regulatory factors affecting the organization's target population. As a provider serving senior citizens in the Denver Metro Area, Ayuda must comply with federal and state requirements for documentation and caregiver credentialing. Familial caregivers who represent a large portion of Ayuda's workforce work under a four-month CNA certification grace period and require continuous tracking to maintain eligibility to ensure patient safety. The current manual system cannot reliably track certification deadlines, nurse visits, or care plan updates, and also cannot efficiently produce audit documentation on demand.

This project is essential to reduce the operational strain on administrative staff, prevent compliance violations, and ensure accurate service delivery to a medically vulnerable population. Evidence gathered during data collection activities include stakeholder interviews, document analysis, observations, and market comparison show inefficiencies, redundant work, bottlenecks in processing approvals, and an inability to consistently maintain accurate records. These findings confirm that a centralized HIPAA compliant digital system is necessary to support patient care, caregiver management, and regulatory reporting. Introducing this system will directly address the organization's most critical weaknesses and improve service quality for seniors who depend on Ayuda's care.

1.4 OBJECTIVE

The primary objective of this project is to design and implement a centralized HIPAA compliant digital information system that replaces Ayuda Home Care's current paper processes and improves the efficiency and accuracy of its core operations. Upon completion the system is expected to enable the organization to manage patient intake, caregiver registration, CNA certification tracking, care plan creation, nurse visit documentation, timesheet processing and compliance reporting through a secure platform.

- Transition from paper records to digital data management, reducing document loss, redundancy, and manual errors.
- Improved tracking of caregiver CNA certification status, including automated alerts for grace period deadlines and expirations.

- Streamlined patient intake and approval workflows, allowing staff to quickly determine whether care is awaiting Medicaid approval, doctor authorization, or caregiver assignment.
- Digitized care plan creation and version control enabling nurses to create, update, and access care plans without reliance on handwritten documents.
- Accurate and efficient timesheet entry supported by staff input and automated logging of corrections or service interruptions.
- Enhanced audit readiness, with the ability to generate complete organized documentation collections for Medicaid and Medicare reviewers.
- Organizational transparency through dashboards that summarize pending tasks, compliance requirements, and operational metrics.

1.5 THE ORGANIZATION

The project team is well suited to design and develop the proposed information system for Ayuda Home Care due to its collective expertise in systems analysis, database design, cybersecurity, and business application development. The team consists of members of academic training and practical experience in information systems, programming, and structured systems development methodologies. Members of the team bring relevant skills in Python, Java, database modeling, and IT troubleshooting in regulated environments and following best practices for access management and secure system operations.

The team includes members with roles such as Systems Analyst who are responsible for gathering requirements, documenting business processes, and ensuring alignment between user needs and technical feasibility. Members also have experience in architecture design and experience in systems analysis, databases, and project management. This supports the team's ability to implement (SDLC). The team operates under a collaborative management structure aligned with standard SDLC governance. Team responsibilities include requirement finding, system modeling, prototype development, documentation, and testing. Each member contributes knowledge to ensure accuracy and quality. Quality control is maintained through iterative review cycles, documentation, and validation of requirements with stakeholders. These strengths position the team to effectively design a secure, reliable, and compliant digital system that meets the operational needs of Ayuda Home Care.

2. PROPOSED TECHNICAL APPROACH

The chosen methodology for the Ayuda Care Information System is Agile. This methodology breaks the process into short iterative sections or sprints which last one to two weeks. The team will maintain frequent communication, often through daily stand-up meetings, to ensure alignment and address issues quickly. Another benefit of using the Agile approach is progress is demonstrated early and regularly, allowing stakeholders to review work throughout the project rather than at the end. This method enables teams to incorporate feedback rapidly and adjust direction when requirements change, helping reduce delays and improve the project's overall effectiveness.

Development Timeline:

Sprint 1: Admin & CNA (Oct 20 - Dec 5)

- Patient and caregiver registration
- CNA certification tracking with automated alerts
- Role-based access control
- Basic compliance dashboard
- Sprint 1 Demo: Dec 5

Sprint 2: Care Plans & Time (Dec 8 - Jan 16)

- Care plan creation interface with service types
- Medicaid and doctor approval workflows
- Nurse visit logging
- Timesheet entry with correction logging
- Service interruption tracking
- Sprint 2 Demo: Jan 16

Sprint 3: Reporting & Compliance (Jan 19 - Feb 20)

- Payroll CSV exports
- Medicaid audit report packages
- Comprehensive dashboards
- System optimization
- Sprint 3 Demo: Feb 20

Stakeholder Interviews (Completed - Aug 18 - Sep 12)

Interviews were conducted with key user groups to understand workflows, pain points, and system expectations:

Administrators currently maintain filing cabinets and prepare payroll from paper timesheets. Problems include misplaced paperwork, time-consuming manual data entry, and no automated deadline alerts. Needs: centralized digital storage, automated compliance alerts, CSV/PDF report generation.

Nurses draft care plans by hand and coordinate with doctors via fax. Problems include difficulty tracking visit history with paper records and time lost on fax coordination. Needs: exclusive care plan editing rights, digital visit logs, electronic document management.

Caregivers submit weekly paper timesheets, but many lack technical skills for online portals. Problems include confusion around CNA certification deadlines and timesheet errors. Needs: office-managed timesheet entry, CNA certification deadline alerts, clear payroll summaries.

Patients and Families call repeatedly to check application status during the two-week Medicaid approval period. Problems include uncertainty about approval timeline and unclear documentation requirements. Needs: streamlined intake process, transparent approval status tracking.

Document Analysis

This analysis revealed high document volume, lack of centralization in physical filing cabinets, standardized data fields suitable for digitization, and compliance documentation requirements for Medicare/Medicaid audits. The team reviewed representative samples of existing operational documents to understand data structures, retention requirements, and approval workflows:

- Paper timesheets (weekly caregiver submissions)
- Faxed doctor approval forms with signatures
- Medicaid approval and denial letters
- Employee background check and E-Verify onboarding forms
- Handwritten nurse care plans and visit notes

Design

The design phase translates requirements into a detailed technical blueprint for the Ayuda Care Information System. The system architecture follows a three-tier model with clear separation between the presentation layer accessed through web browsers, the application layer containing all business logic and workflows, and the data layer managing persistent storage.

Entity-Relationship Modeling

- Normalized to Third Normal Form (3NF)
- Created reference tables to support system configuration and business rules
- Identified core entities

Physical Schema and Data Dictionary

- Created the physical database schema diagram showing all tables and their relationships
- Built a complete data dictionary documenting tables, fields, data types, and examples

Context DFD

- Shows who interacts with the system: patients, caregivers, nurses, doctors, administrators, and government agents
- Maps what goes in (intake forms, timesheets, approvals) and what comes out (reminders, reports, dashboards)

Level-0 DFD

- Breaks the system into seven main processes: registration, care planning, certification tracking, timesheets, reporting, audit logging, and settings
- Shows how data moves between processes and database tables

User Interface Design

- Minimal Clicks: Common tasks require 3-4 clicks maximum
- Progressive Disclosure: Forms reveal fields only when relevant
- Visual Status Indicators: Color-coded badges for certifications, approvals, and alerts
- Responsive Layout: Desktop-optimized with tablet support for field nurses

Prototyping

Created Version 1 prototype demonstrating dashboard, patient registration, and management workflows

- Patient management table with search and filter capabilities
- Built multi-step patient registration form with inline validation and error feedback

Best Practice for Implementation

The development team follows industry-standard practices to ensure code quality, security, and maintainability throughout the project lifecycle. User manuals are created during Sprint 3 covering patient registration, caregiver onboarding, timesheet entry, care plan creation, and report generation to support Ayuda staff training before go-live.

Version Control uses a version control system where code changes are reviewed by another team member before being added to the main codebase. This helps catch errors early and maintains a history of all changes made to the system.

Coding Standards keep the code consistent by following Python best practices such as using clear variable names. Code comments explain complex logic so other team members can understand the workflow. The team writes modular functions that can be called from different parts of the system, reducing redundant code.

Database Management uses a relational database with a normalized schema to reduce redundancy and maintain data integrity. All data is encrypted at rest and in transit. Automated backups are performed daily. Audit logs track all data modifications.

Security Practices follow HIPAA compliance requirements with role-based access control restricting data by user role. Multi-factor authentication is required for all users. Session timeouts protect user accounts. All data is encrypted at rest and in transit to protect patient information

Quality Assurance Plan

The quality assurance strategy ensures that the Ayuda Care Information System meets functional requirements, security standards, and performance expectations before deployment.

Development testing takes place throughout each sprint (October 20 - February 20). This will allow the team to evaluate and verify the functionality of new features before proceeding to the

next phase. To reflect Agile best practices, each team member will test their own code before engaging in peer review. This process will enable the team to identify errors, logic, and workflow issues as they arise. This will reduce reworking in later sprints. For instance, a key workflow such as CNA grace period calculation, or care plan approval can be tested using example data for quality assurance.

User Acceptance Testing will be critical to the deployment phase (March 9-20) when Ayuda's administrative staff and nurses test the system with realistic workflows. Staff will register test patients, onboard caregivers, create care plans, process timesheets, and generate payroll exports. This ensures the system actually works for their daily operations and matches how they need to do their jobs. Feedback from this testing is incorporated before the system goes live.

Performance Testing ensures the system responds quickly under typical workload conditions. The team verifies that dashboards load promptly, and reports generate in reasonable time to support daily operations. Slow database queries or inefficient processes will be identified and optimized before deployment to support smooth daily operations.

Security and Compliance Testing verify HIPAA requirements that are met before deployment. This includes confirming role-based access works correctly, so nurses can only access care plan features while administrators can access the full system. Multi-factor authentication, session timeouts, and data encryption are tested to ensure patient information is protected.

All issues found during testing are logged by the severity level. Critical bugs that prevent core functions like patient registration or timesheet processing are fixed immediately. Less severe issues are prioritized and resolved before deployment. The goal is a stable, usable system that Ayuda staff can rely on from day one.

2.1 REQUIREMENTS

The team has gathered requirements through interviews with Ayuda Home Care's administrative staff and certified nurses. A review of paper documentation was conducted along with existing workflows and direct observations of business processes. Through this process, the team developed a comprehensive understanding of the current challenges that Ayuda faces and the functional expectations and requirements of the proposed information system.

The team found that administrative staff manage many of the agency workflows including patient intake, caregiver onboarding, CNA certification tracking, timesheets entry, and

compliance reporting. Administrative staff rely on paper files and spreadsheets, creating delays, and inconsistent records leading to difficulty preparing documentation for audits. The team concluded that staff need a centralized digital system to reduce manual workload, while improving accuracy and to ensure compliance. The team identified that the certified nurse is solely responsible for creating and updating care plans, recording weekly visits, and communicating with doctors for approval of documents. The current processes rely on handwritten notes and faxed signatures, making version control slow and difficult. The team determined that the system must support digital care plans, visit logs, approval workflows, and controlled editing privileges.

Through interviews and document analysis, the team observed that caregivers do not enter data directly into the system they submit weekly paper timesheets that staff must transcribe. Familial caregivers operate under a four-month CNA grace period the staff must track manually. The team found that the system must support caregiver recordkeeping, grace period management, and accurate hour tracking, even though caregivers themselves will not log in. The team found that patients and families often call the office for updates because staff lack a centralized intake status display. Required documents are stored across multiple paper folders. The team determined that the system must clearly track intake progress, missing documents, approval statuses, and assigned caregivers so staff can efficiently relay information.

Document analysis revealed frequent delays due to fax processes and inconsistent linkage between patient records and provider approvals. The system should record doctor and case manager information, attach approval forms, and track authorization statuses digitally. Medicaid and Medicare audits require assembling large amounts of paper files taking staff days to compile. This risk of incomplete or misplaced documents supports the need for automated audit ready reporting and structured record keeping.

High level functional requirements

Patient Management Requirements	• Digital patient registration records Medicaid ID, contact details, address, doctor, and case manager. • Status tracking across intake, pending approval, active care, interruption, and discharge. • A historical record of updates to maintain auditability and clarity.
Caregiver and CNA Certification Requirements	• Register and manage independent and familial caregivers.

	• Track background checks, credentials, and CNA certification. • Automatically monitor familial caregivers for a four-month grace period and issue alerts as deadlines approach.
Care Plan Requirements	• Exclusive ability of nurse to draft and edit care plans. • Digital workflows for agency approval, doctor approval, and Medicaid authorization. • A version history of all care plan changes.
Nurse Visit Logging Requirements	• Record all patient visits. • Document observations and relevant medical updates. • Receive alerts for overdue visits.
Timesheet and service interruption Requirements	• Input weekly timesheet data submitted on paper. • Capture hours worked per patient and per service. • Tag and record service interruptions. • Maintain detailed audit logs for all corrections.
Reporting and Compliance Requirements	• Payroll ready for CSV exports to assist with accounting. • Automated, audit ready document bundles for Medicaid/ Medicare. • Dashboards that highlight compliance deadlines and missing information.

High Level Non-Functional Requirements

Security and Compliance	• Encryption of data at rest and in transit. • Role based access control and restricted interfaces per user type. • Read only audit logs ensuring integrity and traceability.
Usability	• Interface must be simple and task focused. • Caregivers will not interact with the system directly staff must enter caregiver data.

Performance and Availability	Staff require reliable access during business hours and timely report generation.
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2.2 ARCHITECTURE DESIGN

The Ayuda Care Information System is designed around a three-tier architecture to balance usability, performance, and security. The presentation layer provides a web-based interface accessible through secure login credentials. This interface is optimized for desktop use by administrators and nurses, but it also supports responsive layouts for tablets used during home

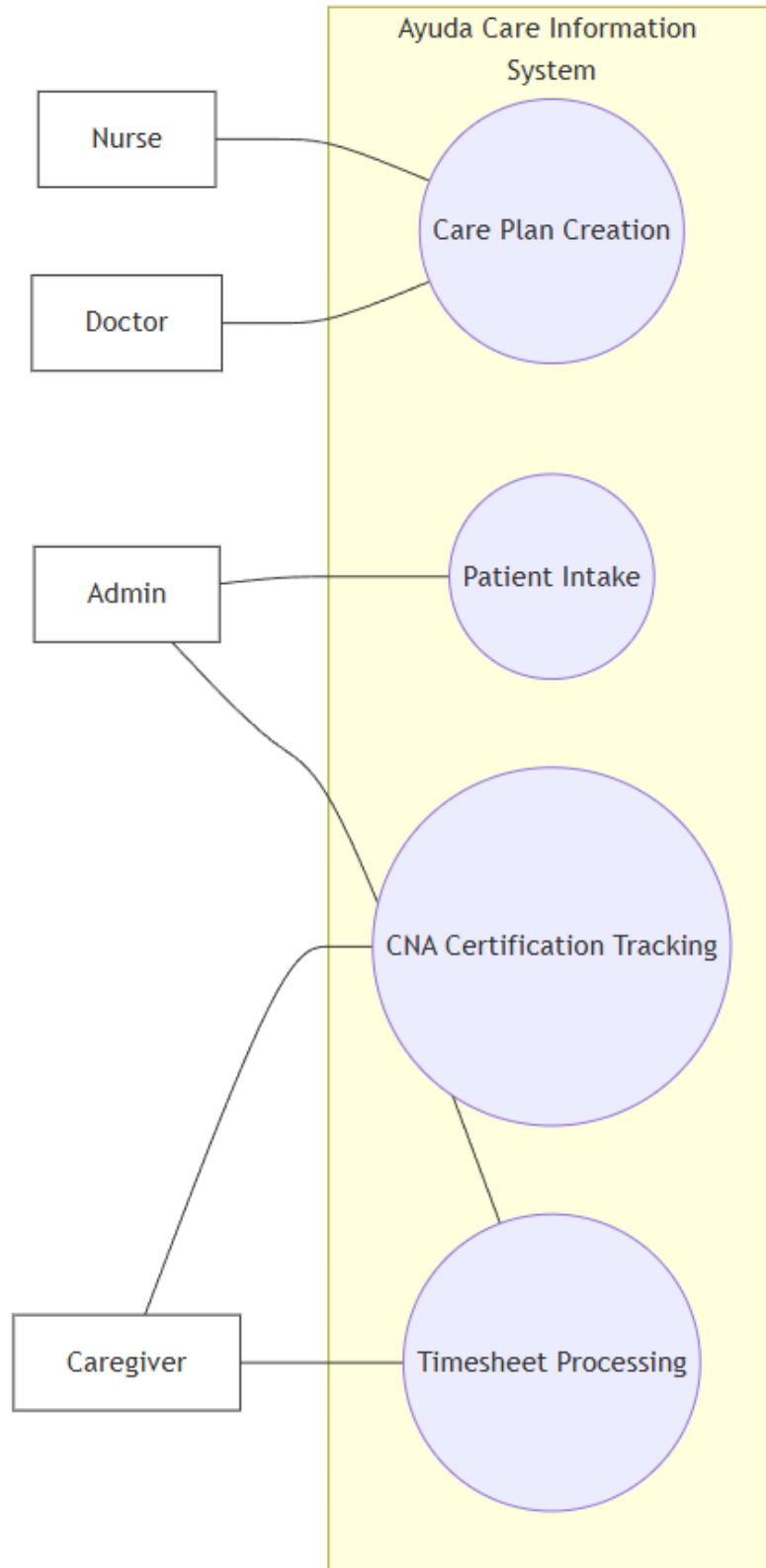
visits. The design focuses on simplicity, with color coded indicators to reduce training needs and minimize errors.

The application layer contains all business workflows. It manages patient intake, caregiver registration, CNA certification tracking, care plan creation, nurse visit documentation, timesheet processing, and compliance reporting. Initially, the prototype was built in Python to validate workflows, but the final implementation will migrate to a more secure and scalable framework such as Django. This ensures stronger authentication, session management, and enterprise-grade security features. The application layer also enforces HIPAA compliance by restricting access based on user roles and logging all actions for audit purposes.

The data layer is managed by a relational database such as PostgreSQL or MySQL, normalized to Third Normal Form to reduce redundancy and maintain integrity. Core entities include patients, caregivers, certifications, care plans, time sheets, and audit reports. Each entity is linked to others to reflect real-world relationships: patients are tied to care plans, caregivers are tied to certifications, and timesheets connect caregivers with patients. Audit reports are generated from these records to support Medicaid and Medicare compliance.

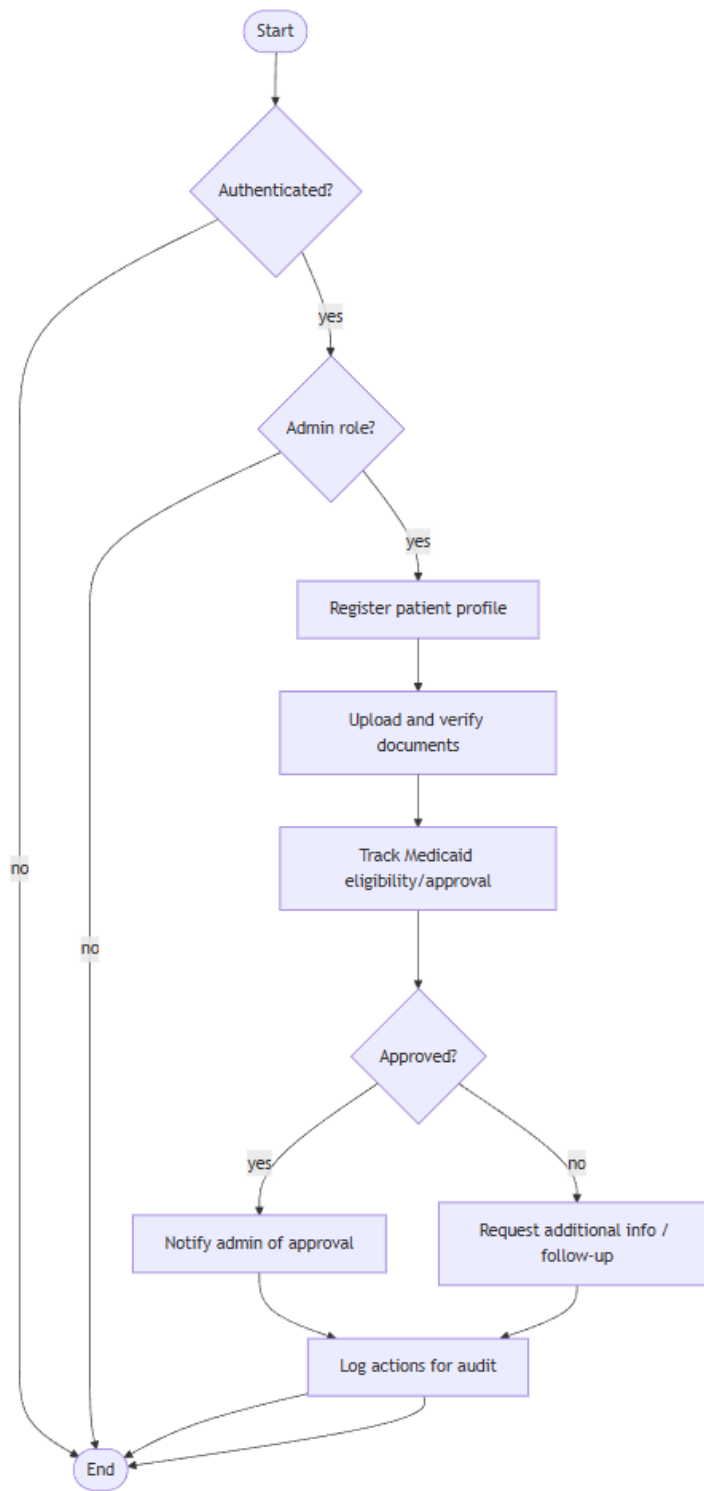
The system will be hosted on HIPAA-compliant cloud infrastructure such as Azure. Hosting will include load balancing to distribute traffic across servers, secure VPN access for remote administrators, and redundant storage with daily backups and disaster recovery protocols. Security is central to the design. All data will be encrypted at rest using AES-256 and in transit using TLS 1.3. Multi-factor authentication will be required for all users, and immutable audit logs will ensure traceability of every change.

Use Case Diagram:

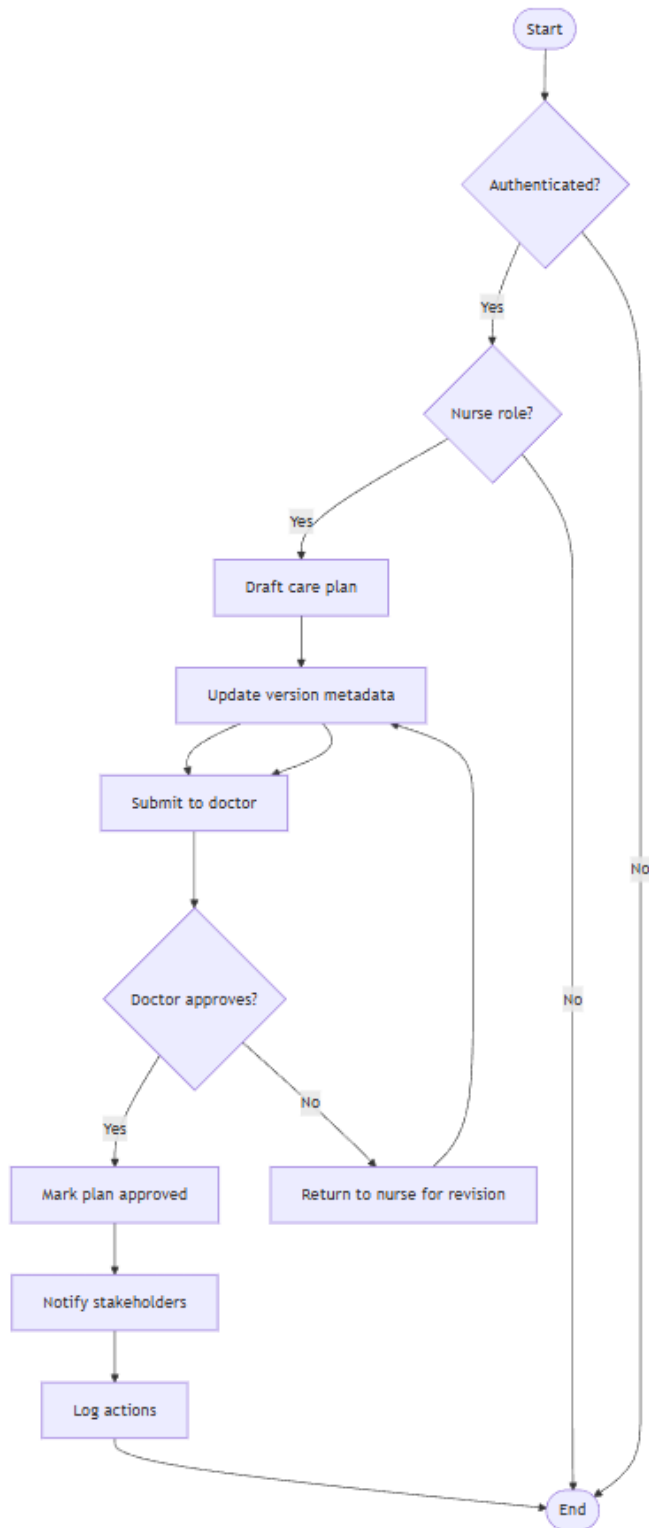


Use Case Narratives:

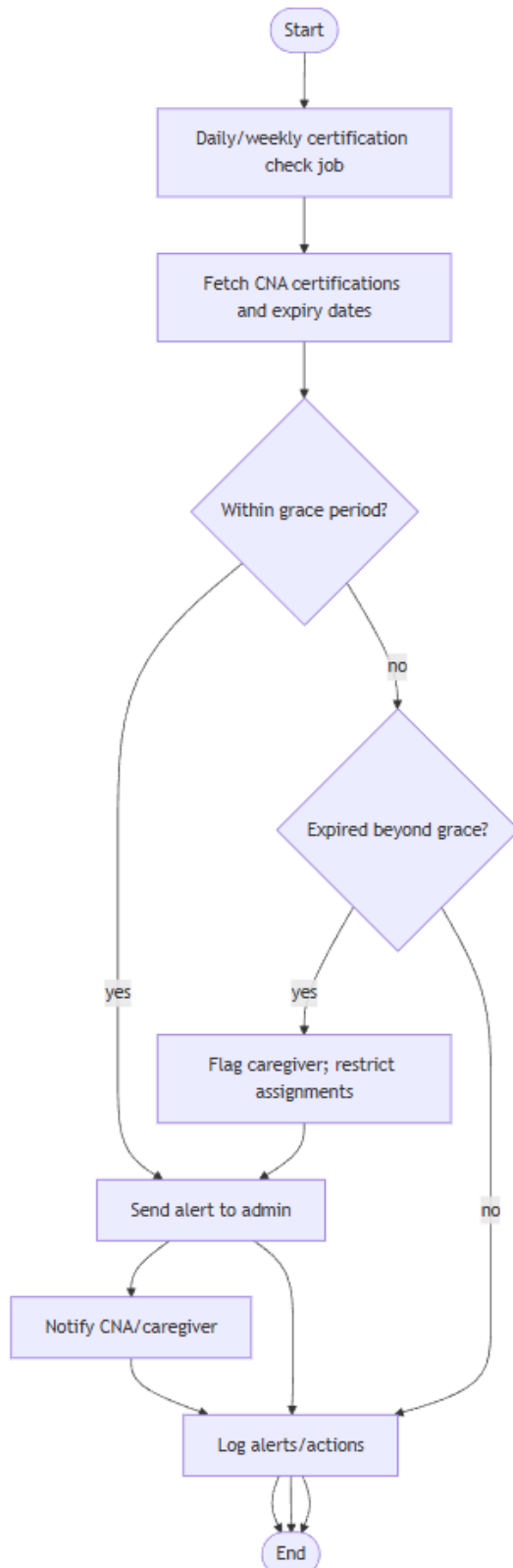
- Patient Intake: Admin registers patient, uploads documents, tracks Medicaid approval.



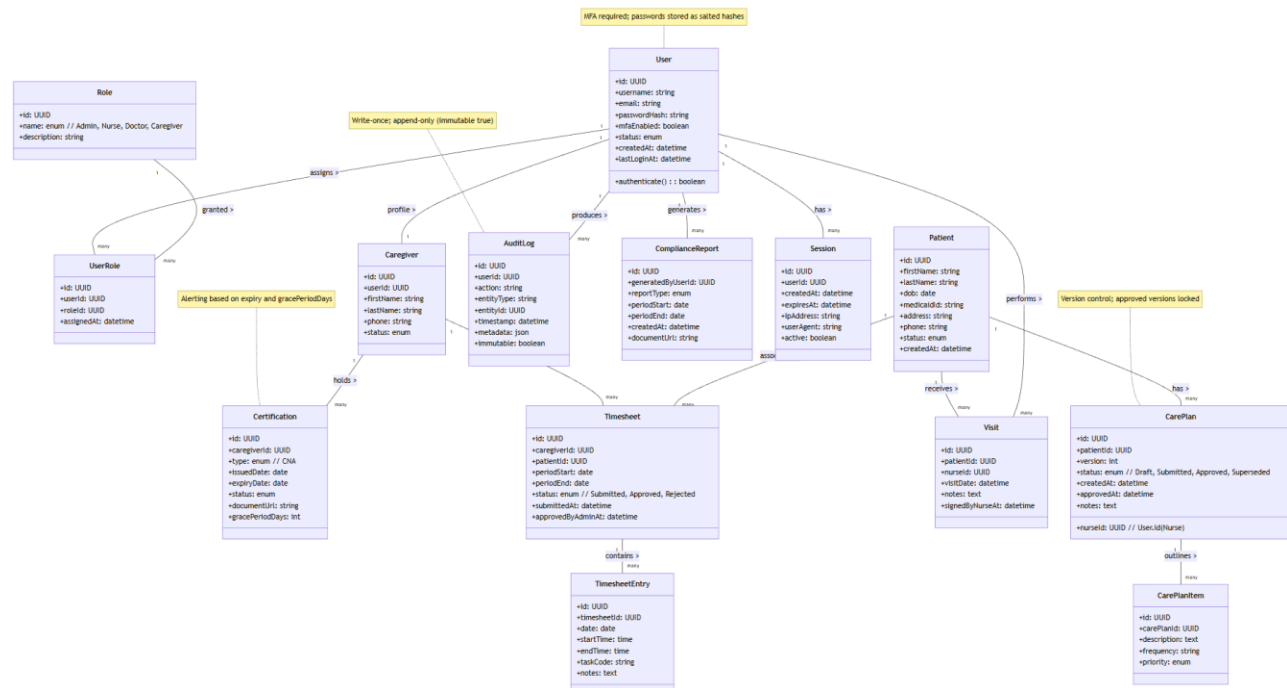
- Care Plan Creation: Nurse drafts plan, submits for doctor approval, version control maintained.



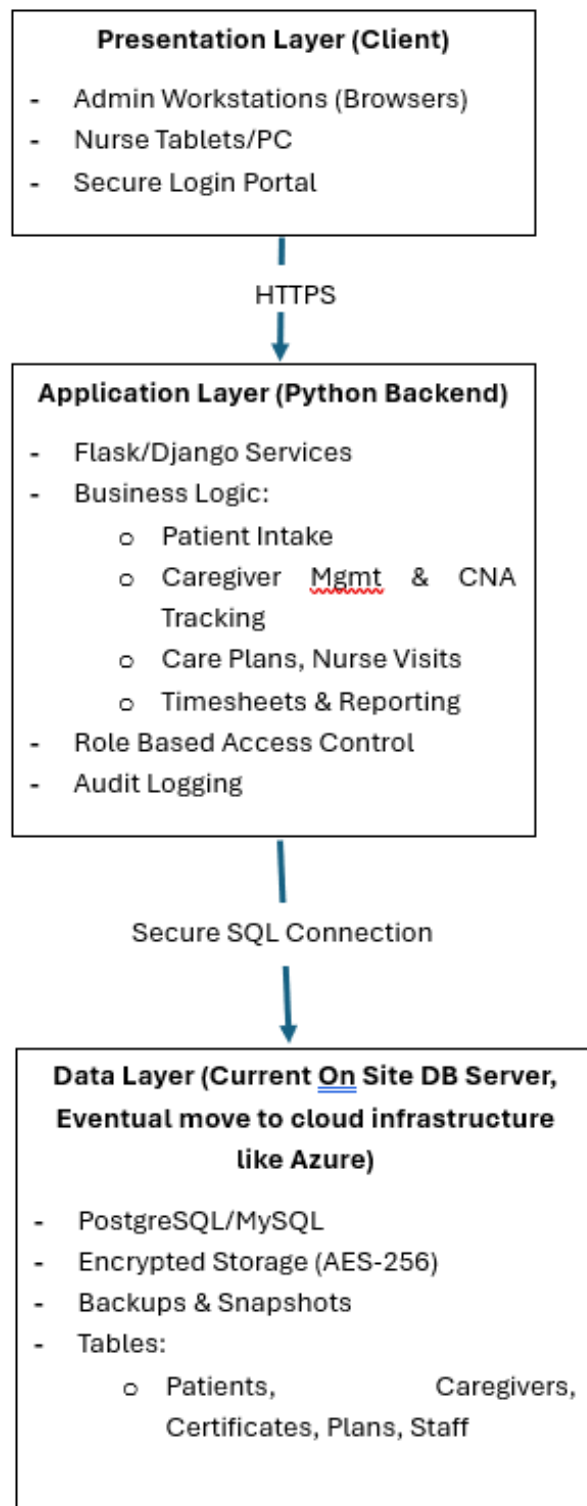
- CNA Certification Tracking: System monitors grace period, sends alerts to admin before expiration.



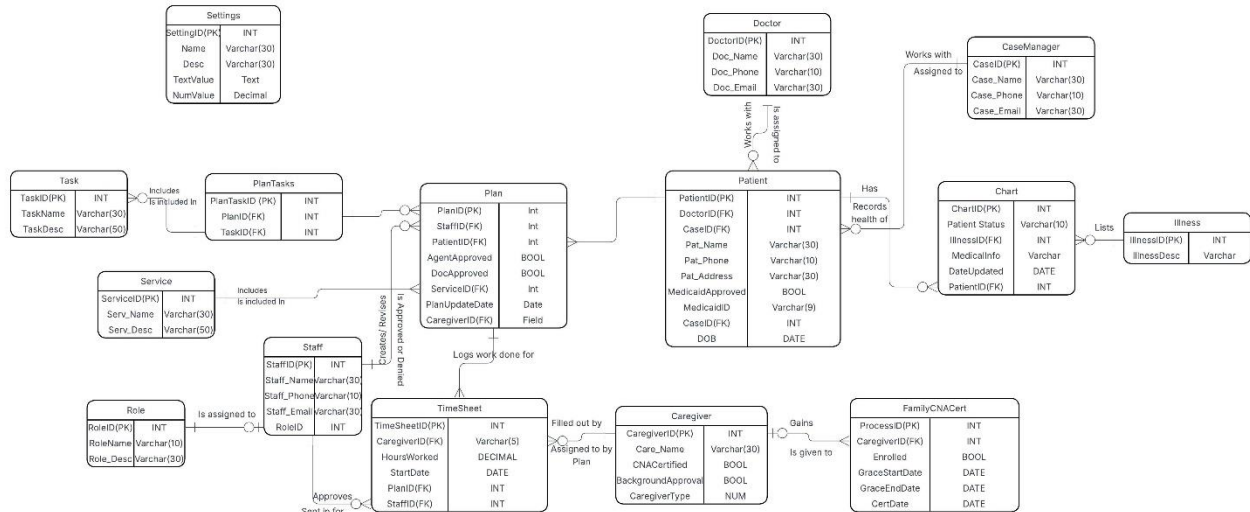
Class diagram:



Architecture Design:



Physical ERD:



2.3 INTERFACE DESIGN

The interface design for the Ayuda Care Information System follows a user approach focused on efficiency. The interface is structured to support the daily operational task of Ayuda's administrative staff and certified nurses while minimizing training complexity and reducing documentation errors.

The design will emphasize task focused navigation with minimal clicks for common actions. Progressive information displays to reduce clutter and guide workflow paths with visual status indicators throughout to alert the user to anomalies and alerts. This approach ensures that users only interact with the features necessary to perform their assigned responsibilities while maintaining full audit traceability.

All users authenticate through a secure login system. The interface loads features based on role permissions. Administrators access patient intake, caregiver management, CNA tracking, timesheets, and reporting. Certified Nurses access assigned patients, care plans, and visit documentation only. This structure enforces HIPAA compliance and prevents unauthorized access to sensitive patients or payroll data. Each system function follows a linear step by step process of selecting the module, then entering or retrieving data, validating required fields and saving a draft or submitting final record. This mirrors administrative and clinical workflows and reduces documentation errors.

The interface is designed so that all common actions require no more than three to four clicks. A left side navigation menu groups system functions by category. Forms display only the fields required for the current step, and additional fields appear only when triggered by system rules

such as status changes or approval workflows. This prevents screen overload and improves usability. Color coded indicators display patient intake status, CNA certification progress, Medicaid and doctor approval status, timesheet approval status. Required fields enforce validation before records can be submitted.

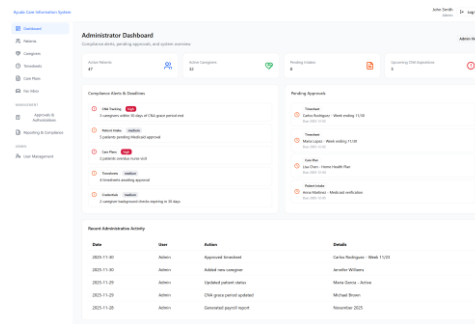
Administrator Workflow

Login – Dashboard

Step 1

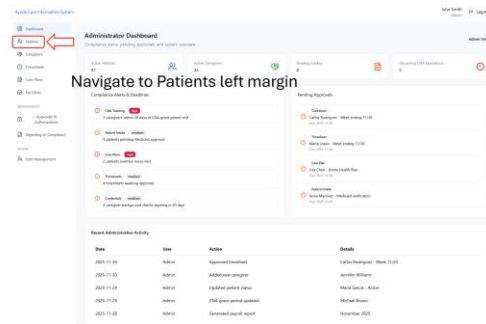


Step 2

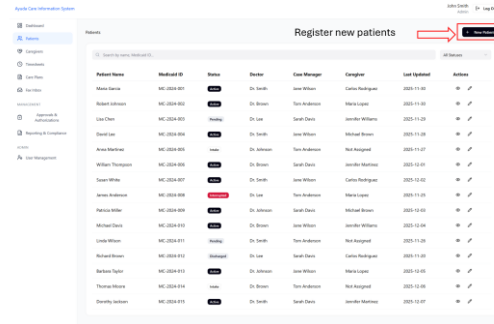


Register Patient – Assign Status – Assign Caregiver

Step 1



Step 2



Step 3

Admin inputs data

Patient Information

First Name: Last Name:

Date of Birth: Gender:

Phone Number: Email:

Address:

City: State: ZIP:

Primary Language:

Emergency Contact

Emergency Name: Relationship:

Phone Number:

Provider Information

Provider Name: Case Manager:

Step 4

Save and create new patient or draft

Patient Information

First Name: Last Name:

Date of Birth: Gender:

Phone Number: Email:

Address:

City: State: ZIP:

Primary Language:

Emergency Contact

Emergency Name: Relationship:

Phone Number:

Provider Information

Provider Name: Case Manager:

Assign Status

Step 1

Name	Medical ID	Status	Gender	Case Manager	Assigner	Last Updated	Action
Robert Johnson	942-2324-841	Active	Male	Dr. Smith	John Williams	2023-11-18	View
Robert Johnson	942-2324-842	Active	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-843	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-844	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-845	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-846	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-847	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-848	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-849	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-850	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-851	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-852	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-853	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-854	Pending	Male	Dr. Smith	John Williams	2023-11-18	View
John Doe	942-2324-855	Pending	Male	Dr. Smith	John Williams	2023-11-18	View

Step 2

Change patient status

Patient Information

First Name: Last Name:

Date of Birth: Gender:

Phone Number: Email:

Address:

City: State: ZIP:

Primary Language:

Emergency Contact

Emergency Name: Relationship:

Phone Number:

Provider Information

Provider Name: Case Manager:

Step 3

Identify reason and submit

Patient Information

First Name: Last Name:

Date of Birth: Gender:

Phone Number: Email:

Address:

City: State: ZIP:

Primary Language:

Emergency Contact

Emergency Name: Relationship:

Phone Number:

Provider Information

Provider Name: Case Manager:

Identify reason and submit

Patient Information

First Name: Last Name:

Date of Birth: Gender:

Phone Number: Email:

Address:

City: State: ZIP:

Primary Language:

Emergency Contact

Emergency Name: Relationship:

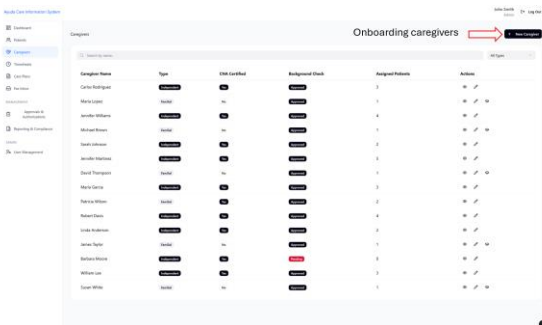
Phone Number:

Provider Information

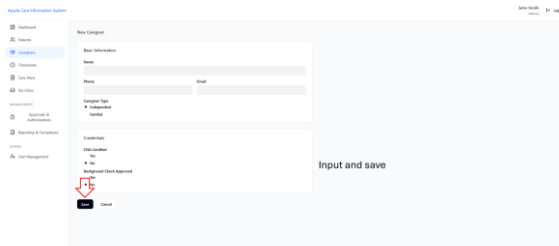
Provider Name: Case Manager:

Register Caregiver – Upload Credentials – Track CNA Grace Period

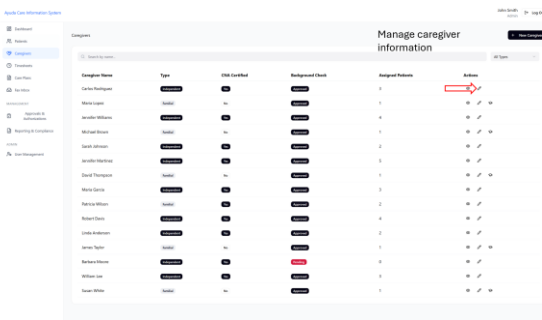
Step 1



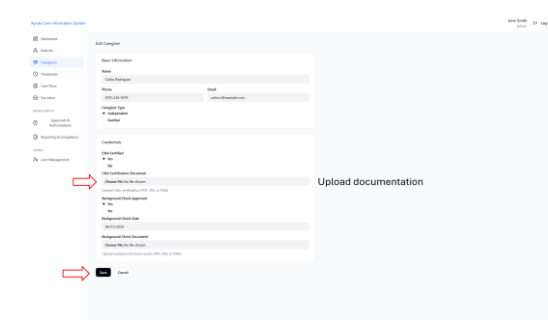
Step 2



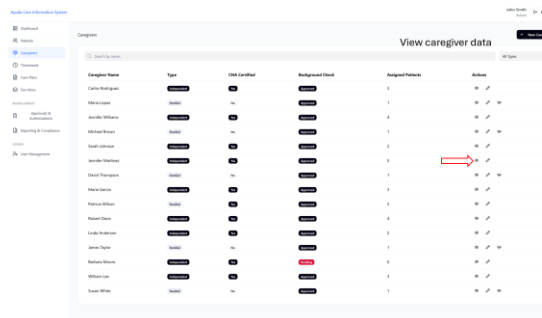
Step 1



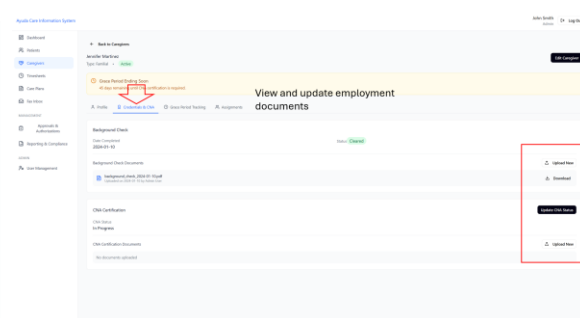
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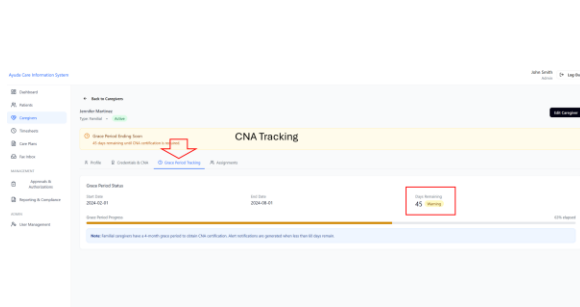
Step 1



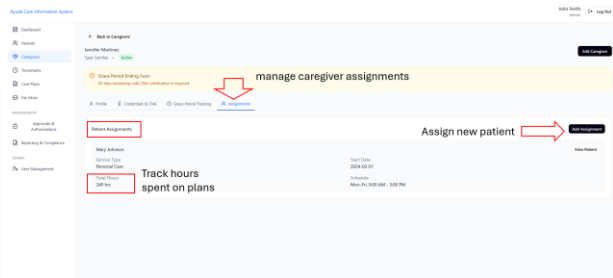
Step 2



Step 3

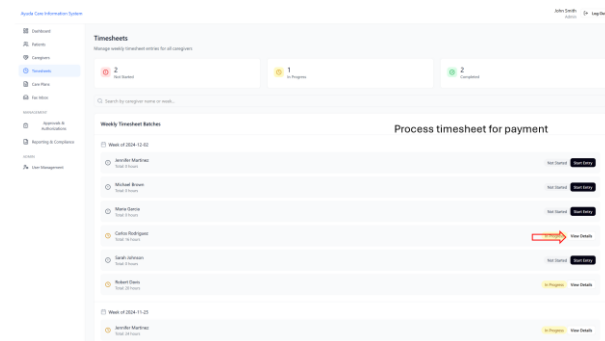


Step 4

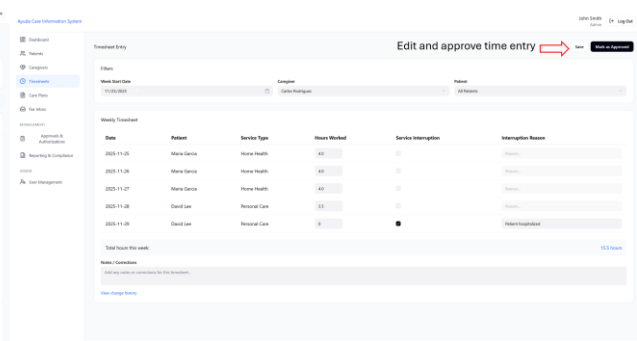


Enter Timesheets – Review – Approve – Generate Payroll Reports

Step 1



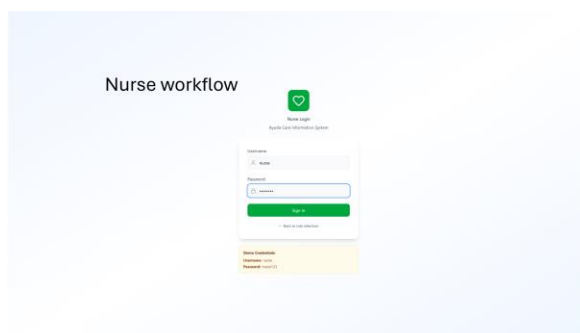
Step 2



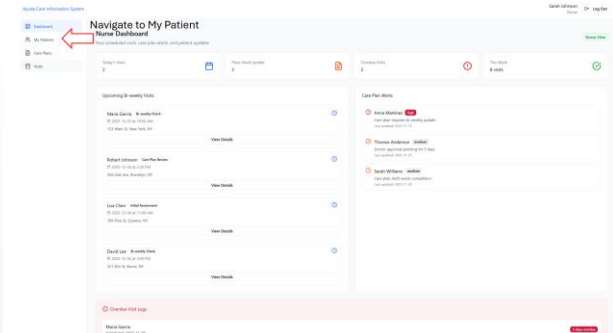
Nurse Workflow

Login – My Patients

Step 1

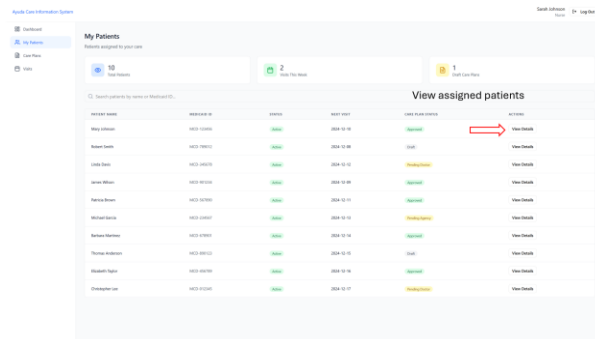


Step 2

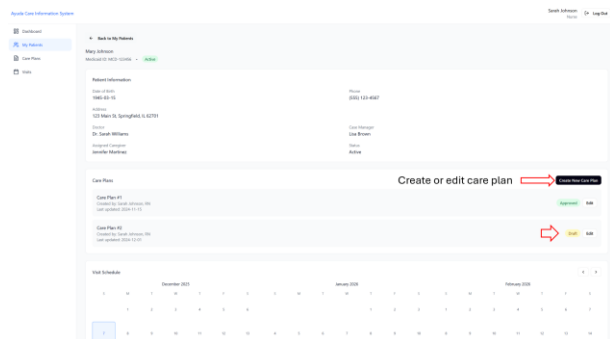


Select Patient – Create or Update Care Plan

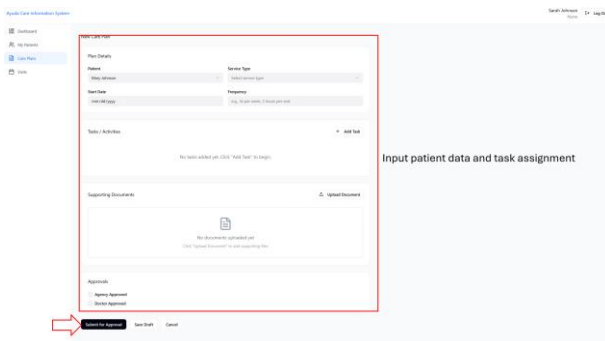
Step 1



Step 2

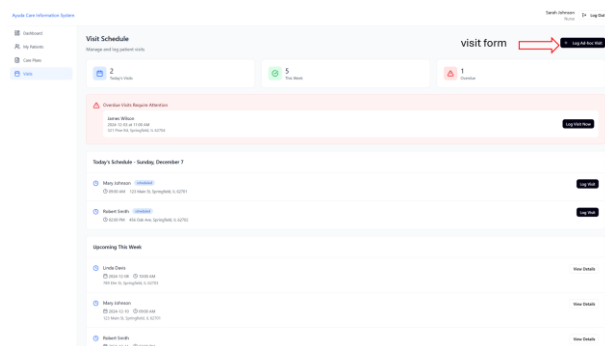


Step 3

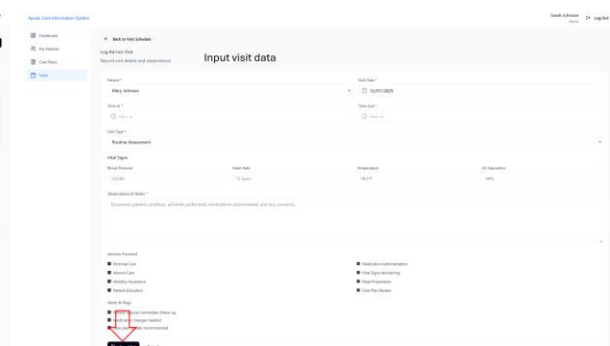


Log Nurse Visit – Submit Observations

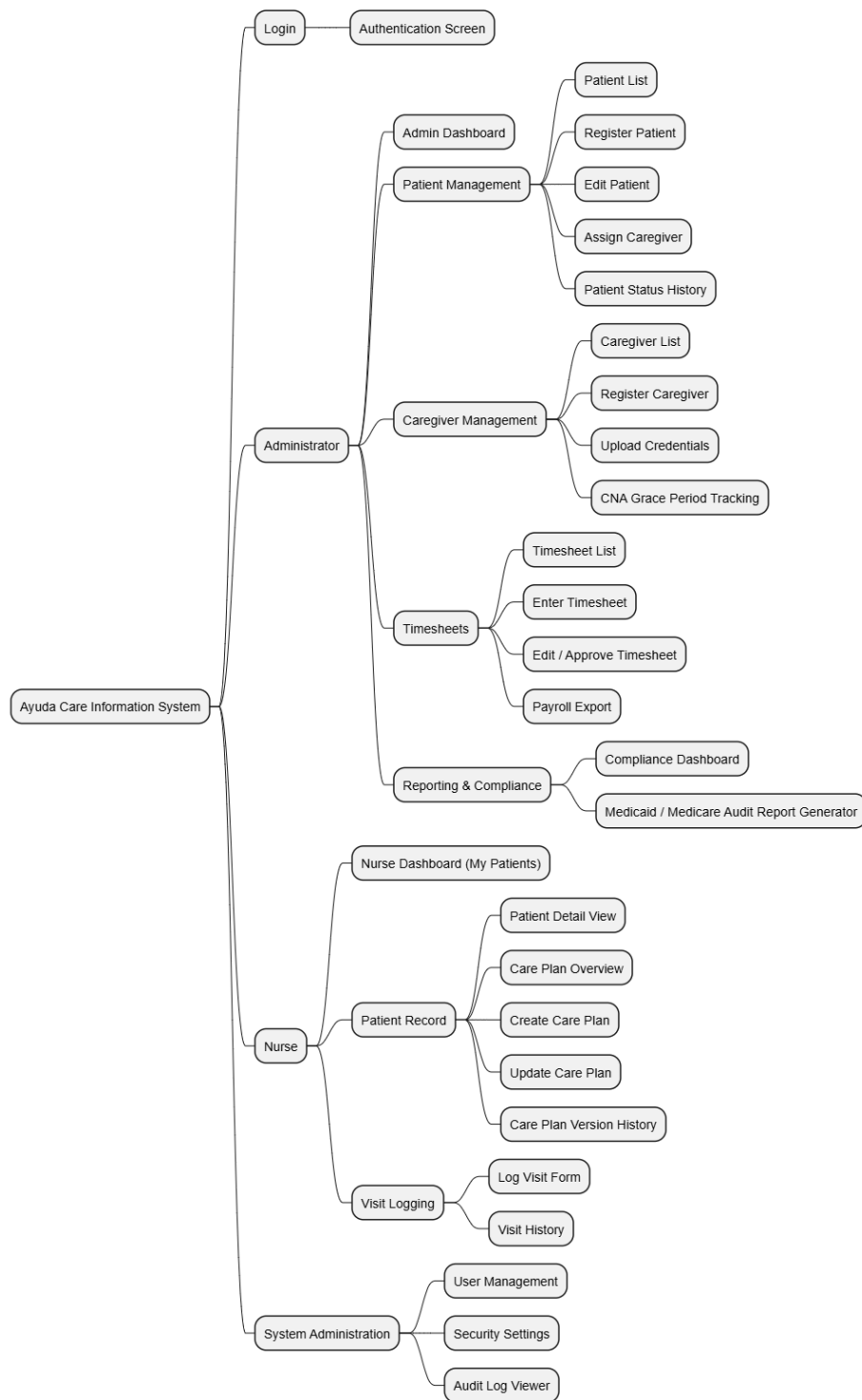
Step 1



Step 2



Ayuda Care Information System Screen Tree:



2.4 QUALITY ASSURANCE PLAN – Testing, future maintenance

The quality assurance plan ensures that the Ayuda Care Information System is reliable, secure, and user-friendly before deployment. Several risks have been identified. Data integrity risks may arise from errors in timesheet transcription or care plan updates. Security risks include unauthorized access or HIPAA violations if role-based permissions are not correctly enforced. Performance risks involve slow dashboard loading or report generation under heavy workloads. Usability risks may occur if staff find workflows confusing or if forms are not intuitive.

Testing will be integrated throughout the Agile development cycle. Developers will perform testing to validate internal logic, database queries, and workflow rules. Black box testing will be conducted by the QA team to confirm that system outputs match requirements without reference to the underlying code. Regression testing will ensure that new features do not disrupt existing workflows. Security testing will include penetration tests, validation of role-based access, and verification of encryption protocols. Performance testing will simulate workloads to confirm that dashboards and reports load within acceptable timeframes, even when multiple users are active.

Quality will be managed through peer code reviews, issue tracking in Jira, and iterative testing embedded in each sprint. Bugs will be logged by severity level, with critical issues resolved immediately and less severe issues prioritized before deployment. User acceptance testing will take place between March 9 and March 20, during which Ayuda's administrative staff and nurses will simulate real workflows such as patient registration, caregiver onboarding, care plan creation, timesheet processing, and payroll export generation. Their feedback will be incorporated before the system goes live.

Maintenance will include scheduled updates, patching, and continuous monitoring for compliance. Success criteria for quality assurance include zero critical bugs at deployment, system uptime of at least 99.9 percent during business hours, audit-ready reports generated within two minutes, and user satisfaction ratings of at least 85 percent during acceptance testing. These measures will ensure that the Ayuda Care Information System is stable, secure, and usable from the first day of operation, supporting both staff efficiency and regulatory compliance.

3. EXPECTED PROJECT RESULTS

The major goal of this project is to deliver a centralized, secure, and digitized information management system tailored to the regulatory and operational needs of Ayuda Home Care Agency. This system means to address the inefficiencies and risks created by Ayuda's current

method of operation by supporting administrative and nursing staff of Ayuda in handling patients, caregivers, and related individuals.

Major Deliverables

1. Centralized Digital Recordkeeping

The system will consolidate patient, caregiver, and other records that will assist Ayuda's Administration in a manner that will reduce the errors from manual or paper entry

2. Patient Care Plan Tools

The tools given to us by the system will allow administrators to track and verify all necessary information and records for the care of patients currently in our system as well as those wishing to enroll.

3. Caregiver Tracking and Reporting Tools

Tools focused on tracking caregiver information and verification allows for an expedient support system of our patients. The capability to track timesheets and certifications streamlines our process in managing patient Care plans.

3.1 MEASURES OF SUCCESS

Centralized Digital Recordkeeping

- Measure: Reduction in documentation errors and misplaced records.
- Current Value: Frequent errors and delays due to paper files and manual entry.
- Target Value: 60–80% reduction in errors through digital forms and centralized records.

Patient Care Plan Tools

- Measure: Faster and more accurate completion of patient intake and care plan updates.
- Current Value: Intake and documentation are often delayed by missing paperwork and unclear statuses.
- Target Value: 30–50% faster processing, with complete, organized care plan histories.

Caregiver Tracking & Reporting Tools

- Measure: Accuracy of caregiver credential tracking and timesheet reporting.

- Current Value: Manual tracking of CNA deadlines and handwritten timesheets create inconsistencies.
- Target Value: 100% on-time CNA certification tracking and 50–70% faster timesheet processing through automated alerts and digital logs.

Overall System Efficiency

- Measure: Staff satisfaction and time saved on administrative tasks.
- Current Value: High workload due to manual entry and repetitive record retrieval.
- Target Value: ≥85% staff satisfaction and significant reduction in administrative hours spent on record management.

4. SCHEDULE

Phase 1 — Planning (Aug 18 – Sep 12) ✓

- 1.1 Kickoff & Governance (Aug 18 – Aug 22) ✓
- 1.2 Requirements Gathering (Aug 25 – Sep 6) ✓
- 1.3 Scope & Sign-off (Sep 8 – Sep 12) ✓

Phase 2 — Design (Sep 15 – Oct 17) ✓

- 2.1 Architecture (Sep 15 – Sep 19) ✓
- 2.2 Data Modeling (Sep 22 – Sep 26) ✓
- 2.3 UX/UI Design (Sep 29 – Oct 3) ✓
- **M2 Design Sign-off (Oct 17)**

Phase 3 — Build (Oct 20 – Feb 20)

- 3.1 Sprint 1: Admin & CNA (Oct 20 – Dec 5)
- **M3 Sprint 1 Demo (Dec 5)**
- 3.2 Sprint 2: Care Plans & Time (Dec 8 – Jan 16)
- **M4 Sprint 2 Demo (Jan 16)**
- 3.3 Sprint 3: Reporting & Compliance (Jan 19 – Feb 20)
- **M5 Sprint 3 Demo (Feb 20)**

Phase 4 — Deploy (Feb 23 – Jun 1)

- 4.1 QA & Bug Fixes (Feb 23 – Mar 6)
- 4.2 UAT & Training (Mar 9 – Mar 20)
- 4.3 Go-Live & Hypercare (Mar 23 – Apr 3)

- **M6 Go-Live (Apr 3)**
- Post-Go-Live Support & Optimization (Apr 6 – Jun 1)

5. BUDGET

The following budget outlines costs for developing, deploying, and maintaining the Ayuda Care Information System over five years. Development labor represents the largest investment at \$441,200, covering three full-stack developers and one project manager working for 200 days to build the complete system including patient management, caregiver tracking, care plan workflows, timesheet processing, and compliance reporting.

Category	Description	Units	Rate/Hour (\$)	Cost (\$)	Notes
DEVELOPMENT LABOR					
Developers (3)	Full-stack development, database design, QA	4800	75	360000	3 devs × 8 hrs/day × 200 days
Project Manager	Project coordination, stakeholder management	1600	50	80000	1 PM × 8 hrs/day × 200 days
Training & Documentation	User manuals, training materials, system docs	40	30	1200	Technical writing
INFRASTRUCTURE					
Hosting (5 years)	Python Web Hosting	60 months		7500	~\$125/month
Integration Costs	Medicare/Medicaid reporting system APIs	One-time		6000	External system setup
OTHER COSTS					
Contingency	Risk buffer for unexpected costs			22735	5% of dev + infrastructure
MAINTENANCE CONTRACT (5 YEARS)					
Annual Maintenance & Support	Bug fixes, security patches, compliance updates, technical support	5 years	10000/year	50000	Mandatory annual contract
5-YEAR TOTAL COST				527435	Ayuda owns all code and IP

Infrastructure costs of \$13,500 include five years of secure web hosting and storage plus one-time integration with Medicare and Medicaid reporting systems. A contingency buffer of 5% (\$22,735) accounts for unexpected costs or scope of changes during development. The mandatory maintenance contract totaling \$50,000 over five years ensures Ayuda receives ongoing bug fixes, security patches, regulatory compliance updates, and technical support to keep the system operational throughout its lifecycle. Upon final payment, Ayuda receives complete source code ownership and full intellectual property rights, providing long-term control and flexibility. Hosting costs continue as operational expenses.

Feasibility Analysis

Technical Feasibility

Equipment - Ayuda's administrative staff will be using existing computer equipment mitigating the need for major hardware upgrades. The application will be hosted on a secure web server. This web-based system will require minimal hardware due to accessibility through standard web browsers. In addition to cost and time savings on equipment installation, staff will benefit from reduced training needs. With reliable network connections, basic computer specifications, and modern browsers, the application will be efficient.

Assessment of current equipment:

Component	Current Specifications	Minimum Required	Status
Desktop Computers (4)	Dell OptiPlex 7080 (or equivalent)	Business-class PC	Sufficient
Operating System	Windows 11 Pro (64-bit)	Windows 10 or newer	Exceeds
RAM (Memory)	8GB DDR4	4GB minimum	Exceeds
Processor	Intel Core i5, 2.9 GHz, quad-core	Dual-core 2.0 GHz	Exceeds
Storage	256GB SSD	50GB available	Exceeds
Monitors (4)	24" LCD, 1920x1080 resolution	1366x768 minimum	Exceeds
Keyboards (4)	Standard USB/wireless keyboards	Standard input device	Sufficient
Mice (4)	Standard USB/wireless optical mice	Standard pointing device	Sufficient
Internet Connection	100 Mbps fiber (business-class)	10 Mbps minimum	Exceeds
Printer (1)	HP LaserJet MFP	Network-capable printer	Sufficient
Document Scanners	Included in MFP	High-speed scanner	Upgrade

Networking	Cisco business router, managed switch	Secure router with firewall	Sufficient
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While existing infrastructure is sufficient for system implementation, the following upgrades are recommended to enhance accessibility and support future growth:

Category	Recommended Action	Cost	Priority	Business Justification
Document Scanners	Purchase 1 high-speed scanner	\$2,000	High	Digitization Efficiency
Dual Monitors	Purchase 4 additional monitors	\$500	Low	Productivity enhancement

These enhancements are not required for initial deployment but should be considered for future acquisition. A dedicated high-speed scanner is a high priority to process ongoing document intake (timesheets, faxes, intake forms), as the existing MFP will remain occupied with daily printing operations. At \$2,000, this represents less than 0.4% of total system investment while directly supporting the digitization of workflows that justify the entire project.

Economic Feasibility

This section evaluates whether the Ayuda Care Information System is financially viable based on estimated costs and expected benefits over a 5-year period. Ayuda will invest an estimated \$527,435, and based on comprehensive evaluation, the project is recommended for approval.

Cost Summary from budget:

- Development Labor: \$441,200 (83.6%) - Largest expense
- Infrastructure (Hosting, 5 years): \$13,500 (2.6%)
- Integration: \$6,000 (1.1%)
- Training & Documentation: \$1,200 (0.2%)
- Contingency (5%): \$22,735 (4.3%)
- Maintenance (5 years): \$50,000 (9.5%)

Total 5-Year Investment: **\$527,435**

Economic Feasibility: The total 5-year investment of \$527,435 is economically justified. Development labor (\$441,200) represents 83.6% of costs, with hosting (\$13,500), maintenance (\$50,000), and contingency (\$22,735) comprising the remainder.

Economic Risks and Benefits

Category	Current Hours/Year (4 staff × hrs/week)	Automated Hours/Year	5-Year Current Cost (@\$30/hr)	5-Year Automated Cost (@\$30/hr)
Manual Data Entry	312 hrs (4 × 1.5 hrs/week × 52)	100 hrs	\$46,800	\$15,000
Audit Preparation	60 hrs (combined staff time)	15 hrs	\$9,000	\$2,250
Total (5 Years)	372 hrs/year	115 hrs/year	\$55,800	\$17,250

Automation will reduce manual data entry and audit preparation by approximately 69%, saving \$38,550 in labor costs over five years. The primary justification is compliance necessity: a single HIPAA violation could cost \$50,000+.

Operational Feasibility: The system significantly improves workflows by transitioning staff from paper-based to centralized digital processes. Staff resistance is mitigated through comprehensive training and gradual rollout. Current staffing (4 administrative employees, 1 nurse) is sufficient with no additional hiring required. Technical risks are managed through the \$10,000/year maintenance contract, daily backups, and parallel operations during transition.

Operational risks and mitigation strategies are summarized below:

Risk	Probability	Impact	Mitigation Strategy
Staff resistance reduces adoption	Medium	High	Comprehensive training, staff involvement in UAT, demonstration of time-saving benefits
Learning curve reduces productivity initially	High	Medium	Parallel operations period, gradual rollout, ongoing support availability
Technical issues disrupt operations	Low	High	\$10K maintenance contract, backup internet connection, daily backups
Caregiver confusion about timesheet process	Low	Low	No change to caregiver workflow, staff handle all system interaction
Data migration errors	Medium	Medium	Phased migration, validation checks, paper backup during transition

Legal Feasibility

This section evaluates whether the Ayuda Care Information System complies with relevant laws and regulations governing healthcare information systems and home care agencies. The system will handle Protected Health Information (PHI) including patient names, Social Security numbers, medical records, and Medicaid identifiers. HIPAA mandates specific safeguards:

Compliance will be achieved through:

- Data encryption in transit and at rest
- Role-based access controls restricting data access to authorized staff only (administrative staff have full system access while nurses are restricted to care plan functionality)
- HIPAA-compliant hosting infrastructure with secure data storage and transmission
- Comprehensive audit logging for all data access

Data Security:

- Encryption of all patient data during transmission and storage
- Secure authentication and session management
- Regular security audits and vulnerability assessments

Workstation and Physical Security:

- BIOS passwords on all administrative computers
- USB ports disabled to prevent unauthorized data transfer
- Automatic screen lock after inactivity
- Encrypted hard drives on all workstations

Operational Security:

- Annual security training for all staff
- Role-based access control limiting data access by job function
- Secure coding practices during development
- Documented security policies and procedures

Backup and Disaster Recovery:

- Daily automated database backups
- Backup retention policy
- Regular backup restoration testing
- Documented recovery procedures

Additionally, the system meets Colorado Medicaid documentation standards including complete care plan records, accurate timesheet documentation, and permanent record retention. Automated alerts for expiring CNA certifications prevent employment of unlicensed caregivers, ensuring ongoing regulatory compliance. The project is legally feasible with no significant barriers to implementation. Legal risks are manageable through secure hosting infrastructure and comprehensive security controls and are lower than current paper-based operations.

6. ANNEXES

Demo Portal: <https://cycle-rise-74433973.figma.site/>